

Ersen'S Joseph

Engineer II – Woods Hole Oceanographic Institution

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Mechanical Engineer specializing in moored platform design, CFD, and hydrodynamic analysis, with experience in subsea mechanical and hydraulic systems, dynamic simulation, and USV dynamics.

TECHNICAL & SOFT SKILLS

Technical	Autodesk Inventor, SolidWorks, ProteusDS, ANSYS Fluent, ABAQUS, COMSOL, MATLAB, Python
Soft Skills	Leadership, highly collaborative team player

EDUCATION

M.S., Mechanical Engineering – <i>University of Massachusetts Dartmouth</i>	Fall 2025
• Thesis: “Mitigation of Vortex-Induced Vibrations in Marine Stretch Hoses Using Helical Grooves”	
(Advisors & Mentor: Dr. Amit Tandon, Dr. J. Thomas Farrar, Donald B. Peters)	
B.S., Mechanical Engineering – <i>University of Massachusetts Dartmouth</i>	May 2022
• Capstone: “Design optimization of an induction coil for drug delivery heating applications”	

PROFESSIONAL EXPERIENCE

Engineer II	March 2026 – Present
Engineer I	May 2023 – March 2026
<i>Woods Hole Oceanographic Institution, Woods Hole, MA</i>	
Production Supervisor	Apr 2018 – May 2022
<i>Mueller Corporation, East Bridgewater, MA</i>	

FIELDWORK & CRUISE PARTICIPATION

F/V <i>Encourager</i> , F/V <i>Voyager</i> – Underwater Acoustic Rainfall Measurement	Apr 2023, Oct 2023, Apr 2024
R/V <i>Tioga</i> , R/V <i>Connecticut</i> , M/V <i>Captain James E Pearce</i> – WFIP-3	Jun – Dec 2024
GNS <i>Aflao</i> – Ocean Margins Initiative (OMI), Ghana	Apr 2025
R/V <i>Thomas G. Thompson</i> – EKAMSAT/ASTRAL, Indian Ocean	May 2025
ROV Jason aboard R/V <i>Atlantis</i> – LIBS system, Pacific Ocean	Jan 2026
Ocean Margins Initiative (OMI) Meteorological Tower, Ghana	Jul 2025, Apr 2026
X-SPAR, South Africa	June 2026

RESEARCH & EDUCATIONAL EXPERIENCE

Graduate Research Assistant – <i>UMass Dartmouth</i>	Sep 2022 – Sep 2025
Undergraduate Research Assistant – <i>UMass Dartmouth</i>	Jan 2021 – May 2022
Teaching Assistant - <i>UMass Dartmouth</i>	Fall 2022
Intern, Blue Economy Program - <i>UMass Dartmouth - WHOI</i>	Spring 2022

PRESENTATIONS

Marine Technology Society Buoy Workshop (Talks), 2024, 2026
MUST Day (Posters), 2024, 2025
UMass Dartmouth Mechanical Engineering Seminar (Talks), 2023, 2025
Sigma Xi Research Exhibition (Poster), 2022
Massachusetts Undergraduate Research Conference (Poster & Talk), 2022

CERTIFICATIONS

Fluid Dynamics Simulation Using ANSYS – Cornell University (2023)
Responsible Conduct of Research for Engineers – CITI (2020)

HONORS & AWARDS

NASA Group Achievement Award – S-MODE (2023)